

NAME: _____

PERIOD: _____

Getting to School: Survey Data

AP Statistics · Unit 1 · Topic 1.2

[STAR] THE PROBLEM

A school newspaper surveyed six students for a story about getting to school. The table records lunch period, ZIP code, commute time in minutes, and a 1–5-star opinion rating of the commute.

Two reporters disagree about the numbers in it. **Maya:** “Zip code and Rating are both quantitative — they’re numbers, so we can average them.” **Jordan:** “They’re both categorical. Numbers can be labels.”

Getting-to-school survey (minutes; 1–5 stars)

Name	Lunch	ZIP	Min	Stars
Aiden	A	01902	12	4
Bea	B	01905	35	2
Cruz	A	01902	8	5
Dev	C	01904	22	3
Elise	B	01905	41	1
Fiona	A	01901	15	4

FIRST TAKE — one sentence before you work the parts

Would averaging two ZIP codes tell a reader anything useful about getting to school? Give one reason.

[BOOK] VARIABLES (keep on the page)

An **individual** is a person or thing the data describe. A **variable** is a characteristic that can change between individuals. An identifier only names an individual. **Categorical** values are category names or group labels, even when written with digits. **Quantitative** values are numbers for a measured or counted quantity. State units for measurements. Before averaging, ask what quantity the values measure or count.

Work (a)–(c).

DOK 1 (a) Who are the individuals in this data set?

DOK 2 (b) Classify Lunch, ZIP, and Min as categorical or quantitative. Give the unit for Min.

DOK 3 (c) Decide whose ZIP-code claim is right and explain why an average ZIP code would or would not be meaningful. Then defend a classification for the star rating using what the numbers represent.

Frame: I agree with _____ about ZIP because those numbers represent _____. An average would not tell us _____.

Frame: I treat the star rating as _____ because I assume the numbers mean _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored **E / P / I**.

Optional #1 — not collected

DOK 2 For each variable, write **C** (categorical) or **Q** (quantitative); for each Q, give the unit: (1) jersey number of a soccer player; (2) number of goals scored this season; (3) shoe size; (4) preferred position (goalkeeper, defender, midfielder, forward); (5) minutes played in the last match.